Speed and Density Planning for a Speed-Constrained Robot Swarm Through a Virtual Tube

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***Abstract*—The planning and control of a robot swarm in a complex environment has attracted increasing attention. To this end, the concept of virtual tubes has been taken up in our previous work. Specifically, a variable-width virtual tube is designed to prevent collisions with obstacles in complex environments. Based on the planned virtual tube for a large number of speed-constrained robots, the average forward speed and density along the virtual tube are further planned in this letter to improve safety and efficiency. In contrast to existing methods, the proposed method leverages global information and is well-suited for navigating confined spaces with speed-constrained robot swarms. Numerical simulations and experiments are conducted to show that the safety and efficiency of the passing-through process are improved.**

***Index Terms*—Swarm robotics, constrained motion planning, motion control, virtual tubes.**

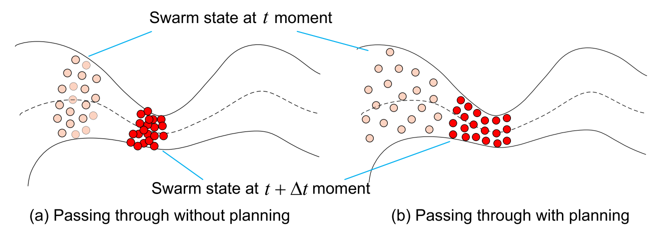

Fig. 1. Comparison of the swarm distribution in a virtual tube without planning and with planning. Light red indicates the swarm distribution before dark red.

I. Introduction

SWARM planning and control in complex environments has garnered increasing attention. The primary objective is to identify an optimal route for each robot from the starting point to the goal subject to the kinematic conditions, ensuring that no collisions occur with other robots or obstacles. Determining how to enable robot swarms to pass through complex environments more safely and efficiently is a critical issue that researchers are actively exploring [1].

Many methods have been proposed for the passing-through problem of the robot swarm in a complex environment, such as formation control [2], [3], multi-robot trajectory planning [4], [5] and control-based methods [6]. Considering the scalability of the control-based method at the swarm scale and its applicability to decentralized control, control within a virtual tube is proposed [7], [8], where all robots sharing one planned virtual tube are under distributed control. The virtual tube can be viewed as a safety corridor, in which the presence of obstacles is precluded.

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A video about the simulations and experiments is available at https://youtu.be/F3Xg1vUcxws.

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Nevertheless, when attempting to direct a multitude of robots through some confined spaces within a complex environment, these methods may encounter inherent constraints. In this case, the robustness and scalability of formation control exhibit constraints, and the calculation amount of multi-robot trajectory planning increases dramatically, which also depends on direct communication heavily. Moreover, control-based methods easily lead to congestion. With regard to the virtual tube control method, narrowing virtual tubes generated based on narrow spaces will result in congestion during the large-scale swarm passing-through process, which slows down the speed of the swarm. For speed-constrained robots, such as fixed-wing unmanned aerial vehicles (UAVs), the swarm is unable to stop to avoid colliding with each other when entering the narrowing virtual tube based on our previous control method [7]. This poses serious safety risks as shown in Fig. 1(a).

The approaches for swarm navigation in confined spaces can be broadly categorized into two main categories. The first involves the utilization of micro-robots with limited physical dimensions [9]. The second employs a formation-based strategy, leveraging variations in formation and alterations in communication topologies to pass through narrow spaces [10]. However, these methods are constrained by factors such as the shape of the formation and communication dependencies, rendering them less versatile and adaptable across diverse scenarios. Additionally, enabling a speed-constrained swarm to navigate complex environments without collisions presents significant challenges. To address this, a decentralized flocking policy with obstacle avoidance is developed for multiple fixed-wing UAVs utilizing a multi-agent deep reinforcement learning approach [11].

The planning-based approach represents an attractive strategy to address the challenges posed by confined spaces and speed limitations. For example, speed planning is primarily designed to optimize the speed of robots as well as save energy [12]. The main idea of speed planning is to provide a path and speed planner under the physical constraints of the robot [13]. In addition, density planning is an effective method to improve the safety of the swarm. For instance, a density planner is designed to generate a trajectory with the minimum collision probability under dynamic obstacles based on the initial distribution [14]. Furthermore, the optimal control problem is solved to make the swarm quickly converge to the desired density distribution [15].

This letter presents the speed and density planning as well as traversing control for a speed-constrained robot swarm based on the established narrowing virtual tube to pass through some narrow spaces. Density planning is employed to determine the optimal density at various positions along the tube, thereby mitigating the risk of collisions due to the inability to stop as shown in Fig. 1(b). Simultaneously, the average forward speed along the tube is optimized to enhance overall efficiency. Subsequently, distributed control is implemented for each individual in the swarm to track the planning results. Compared to previous implementations of the virtual tube, enhancements have been made to the guidance vector field within the virtual tube. Furthermore, a density scalar field within the virtual tube is designed for the swarm. The proposed method improves efficiency on the basis of safety, and extends the applicability of the virtual-tube-based control method to more application objects and more obstacle-dense scenes, particularly those involving slits.

The contributions of this letter are as follows.

- A hierarchical approach is proposed that first performs speed and density planning, then distributed control to track the planning results, which improves safety and efficiency through trade-offs. It addresses the issue of a speed-limited swarm passing through a narrowing virtual tube, which extends the applicability of virtual tubes to more complex scenarios containing slits.
- The statistical results obtained through extensive comparative simulations with other state-of-the-art methods demonstrate the superiority of this approach. Furthermore, the corresponding applications on cars and quadcopters are verified in experiments.

II. Problem Formulation

For simplicity of explanation, the problem is presented in two dimensions in this letter. However, the methodology can be seamlessly extended to three dimensions.

A. Robot Modeling

1) Robot Kinematic Model: A robot is represented as a two-dimensional mass point model. The swarm is composed of N homogeneous robots. In the Cartesian coordinate system, the motion model of the ith robot is

$$\dot{\mathbf{p}}_i = \mathbf{v}_{\mathrm{c},i}, \tag{1}$$

where $i = 1, 2, \ldots, N$, $\mathbf{p}_i \in \mathbb{R}^2$ represents the position of the ith robot, $\mathbf{v}_{\mathrm{c},i} \in \mathbb{R}^2$ represents the velocity command of the ith robot.

In accordance with the limitations of robot mobility, the robots are restricted by the maximum speed $v_{\max}$, the minimum speed $v_{\min}$, the maximum tangential acceleration a_{v}, and the maximum normal acceleration a_{n} as follows:

$$0 < v_{\min} \leq \|\mathbf{v}_{\mathrm{c},i}\| \leq v_{\max}, \tag{2}$$

$$\mathrm{d}\,\|\mathbf{v}_{\mathrm{c},i}\| / \mathrm{d}t \leq a_{\mathrm{v}}, \tag{3}$$

$$\|\mathbf{v}_{\mathrm{c},i}\|^2 / r_{\mathrm{t}} \leq a_{\mathrm{n}}, \tag{4}$$

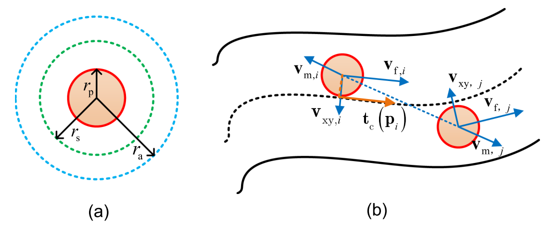


Fig. 2. (a) The physical area, safety area, obstacle avoidance area of a robot. (b) The velocity command of the ith robot and the jth robot.

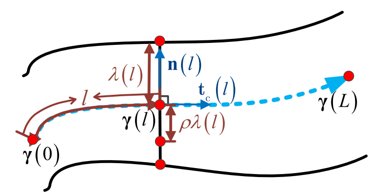


Fig. 3. Schematic diagram of a virtual tube. The black solid curves represent the boundaries of the virtual tube, the light blue dashed line represents the generating curve of the virtual tube, and the red solid point represents some special points.

where $r_{\mathrm{t}} > 0$ represents the radius of the robot's trajectory curvature.

2) Physical Area, Safety Area and Obstacle Avoidance Area of Robots: Concentric circles of varying sizes are used to represent the physical area, safety area, and obstacle avoidance area of robots. As shown in Fig. 2(a), $r_{\mathrm{p}}, r_{\mathrm{s}}, r_{\mathrm{a}}$ denote the radius of the physical area, the safety area, and the obstacle avoidance area respectively. Besides, there exists $r_{\mathrm{p}} \leq r_{\mathrm{s}} \leq r_{\mathrm{a}}$ [7]. Particularly, r_{a} is a controlled variable corresponding to the planned swarm density in this letter.

B. Virtual Tube Modeling

A virtual tube is a *regular* curved tube designed on a two-dimensional plane [16]. As illustrated in Fig. 3, the virtual tube in a two-dimensional plane is expressed as:

$$\mathcal{J}(l, \theta, \rho) = \boldsymbol{\gamma}(l) + \rho\lambda(l)\mathbf{n}(l)\cos\theta, \tag{5}$$

in which $\theta = \{0, \pi\}$, $l \in [0, L]$, $\rho \in [0, 1]$. The curve $\boldsymbol{\gamma}(l)$ is the *generating curve* of the virtual tube. The vector $\mathbf{n}(l)$ represents the normal vector of the generating curve, $\lambda(l)$ is a continuous real-valued function with respect to l, which represents the widths of the virtual tube. l represents the arc length of the generating curve from the starting point $\boldsymbol{\gamma}(0)$, and $\boldsymbol{\gamma}(l)$ is the position with the arc length l along the generating curve from $\boldsymbol{\gamma}(0)$, θ represents the left or right half of the virtual tube divided by generating curve, ρ represents the ratio of the distance between a specific point within the virtual tube and the corresponding point on the generating curve $\boldsymbol{\gamma}(l)$ to the corresponding radius $\lambda(l)$. $L > 0$ represents the whole length of the generating curve. The term $\mathcal{J}(l, \theta, 1)$ represents the boundaries of the virtual tube. Additionally, $r_{\mathrm{t}}(l)$ represents the curvature radius of the tube generating curve. The generation of the regular virtual tube initiates with a path planning algorithm that produces a generating curve. Subsequently, the regular virtual tube is established by resolving an optimization problem subject to specific constraints. The detailed theories and methods of virtual tube generation are presented in our previous work [16], [17]. In this letter, the position of the ith robot within the virtual tube is defined as $\mathbf{p}_i = \mathcal{J}(l_i, \theta_i, \rho_i)$. Particularly, each $\mathbf{p}_i$ corresponds to a unique l_i, where $l_i \in [0, L]$.

C. Robot Controller

In this letter, the movement of the robot is controlled by velocity command. As shown in Fig. 2(b), the velocity command $\mathbf{v}_{\mathrm{c},i}$ of the ith robot is comprised of three components: $\mathbf{v}_{\mathrm{f},i}$ to guide the robot to move forward along the virtual tube, $\mathbf{v}_{\mathrm{m},i}$ to prevent the conflict among robots, and $\mathbf{v}_{\mathrm{xy},i}$ to restrict the robot in the virtual tube. Particularly, $\|\mathbf{v}_{\mathrm{c},i}\| \neq 0$. Our previous work [7] describes the controller design in detail. Here it is omitted for space limitation. For the ith robot, the velocity command is

$$\mathbf{v}_{\mathrm{c},i} = \mathrm{sat}\left(\underbrace{v_{\mathrm{f},i}\mathbf{t}_{\mathrm{c}}(\mathbf{p}_i)}_{\mathbf{v}_{\mathrm{f},i}} + \underbrace{\sum_{j \in \mathcal{N}_{\mathrm{m}.i}}^{N} b_{ij}\tilde{\mathbf{p}}_{ij}}_{\mathbf{v}_{\mathrm{m},i}(r_{\mathrm{a},i})} + \underbrace{c_i \mathbf{A}_{\mathrm{xy}} \tilde{\mathbf{p}}_{\mathrm{xy},i}}_{\mathbf{v}_{\mathrm{xy},i}(r_{\mathrm{a},i})}, v_{\min}, v_{\max} \right)$$

with

$$\mathrm{sat}(\mathbf{v}, v_{\min}, v_{\max}) \triangleq \begin{cases} v_{\min}\mathbf{v}/\|\mathbf{v}\| & \|\mathbf{v}\| < v_{\min} \\ \mathbf{v} & v_{\min} \le \|\mathbf{v}\| \le v_{\max} \\ v_{\max}\mathbf{v}/\|\mathbf{v}\| & \|\mathbf{v}\| > v_{\max} \end{cases}, \tag{6}$$

where $b_{ij} > 0, c_i > 0$ are the coefficients, $v_{\mathrm{f},i}$ represents the modulus of $\mathbf{v}_{\mathrm{f},i}$, and $\mathbf{t}_{\mathrm{c}}(\mathbf{p}_i)$ represents the tangent vector of the projection point of the ith robot on the virtual tube generating curve, as depicted in Fig. 2(b). Vector $\tilde{\mathbf{p}}_{ij} = \mathbf{p}_i - \mathbf{p}_j$ represents the position error between the ith robot and the jth robot, $\mathcal{N}_{\mathrm{m}.i}$ represents the set of robots in the obstacle avoidance area of the ith robot. Vector $\tilde{\mathbf{p}}_{\mathrm{xy},i} = \mathbf{p}_i - \mathbf{p}_{\mathrm{t},i}$ represents the position error between the ith robot and the boundary of the virtual tube, in which $\mathbf{p}_{\mathrm{t}}$ is defined as

$$\mathbf{p}_{\mathrm{t},i} = \arg\min_{\mathbf{p}_{\mathrm{t}}} \|\mathbf{p}_i - \mathbf{p}_{\mathrm{t}}\|, \mathbf{p}_{\mathrm{t}} \in \mathcal{J}(l, \theta, 1). \tag{7}$$

The matrix $\mathbf{A}_{\mathrm{xy}}$ is a projection matrix that projects $\tilde{\mathbf{p}}_{\mathrm{xy},i}$ into the direction perpendicular to $\mathbf{v}_{\mathrm{f},i}$. Particularly, $\|\mathbf{v}_{\mathrm{m},i}(r_{\mathrm{a},i})\|$ and $\|\mathbf{v}_{\mathrm{xy},i}(r_{\mathrm{a},i})\|$ will increase at the moment when $r_{\mathrm{a},i}$ is increased. In other words, the swarm will expand similarly to the expansion of gas upon heating. In addition to incorporating the minimum speed constraint, the form of the controller remains identical to that in paper [7].

When $\mathbf{v}_{\mathrm{f},i} + \mathbf{v}_{\mathrm{m},i} + \mathbf{v}_{\mathrm{xy},i} = \mathbf{0}$, the directional orientation of the velocity command is preserved, and its magnitude is constrained to $v_{\min}$ by the saturation function delineated in (6). Given that $v_{\min} > 0$ in (2), it is theoretically untenable for the controller to guarantee safe traversal. This implies the potential occurrence of collisions among robots and the robot surpassing the boundaries of the virtual tube. Nevertheless, the density planning proposed in this letter serves as a substantial enhancement to the safety of swarm passage. Furthermore, due to the saturation limits of the controller and the employment of a single integrator model, the direction of velocity may exhibit jitter in practical applications, which will be further studied in future research.

Remark 1: When a robot is represented as a single integrator such as (1), exemplified by some holonomic kinematics robots such as multicopters, helicopters, and omni-directional wheeled robots, $\mathbf{v}_{\mathrm{c},i}$ can be straightforwardly employed to control the robot. When dealing with a more complicated model, such as a second-order integrator model, additional control laws become imperative. In our previous work [7], we introduced a *filtered position model* that transforms a second-order model into a first-order model just like (1). As for certain nonholonomic kinematics robots such as ground mobile robots and fixed-wing UAVs, it is possible to generate an appropriate forward speed command or angular speed command tailored to the model. This process ensures that the robot velocity can track the designed velocity command $\mathbf{v}_{\mathrm{c},i}$, that is, $\lim_{t\to\infty}\|\mathbf{v}_i(t) - \mathbf{v}_{\mathrm{c},i}\| = 0$ [18]. An alternative approach involves the utilization of a near-identity diffeomorphism to establish a connection between the desired single integrator model and the more precise robot model [19].

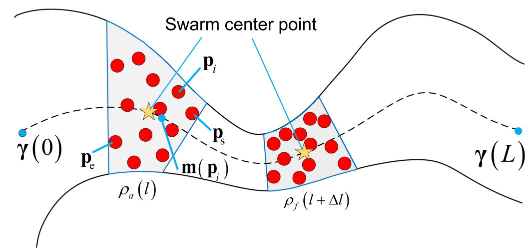


Fig. 4. Distributions of the swarm in the virtual tube at the position with arc length l and $(l + \Delta l)$.

D. Density and Average Forward Speed

In this letter, the swarm density ρ_{a} is defined as the number of robots in a unit area,

$$\rho_{\mathrm{a}} = n/S, \tag{8}$$

where S is the area occupied by n robots within the virtual tube. In the following, take $n = N$ as an example to illustrate the method for calculating the area S occupied within the virtual tube, as the gray area in Fig. 4. Assume that the swarm passes through the virtual tube from the starting point $\boldsymbol{\gamma}(0)$ to the ending point $\boldsymbol{\gamma}(L)$. Let $\mathbf{p}_{\mathrm{e}}$ denote the position of the robot farthest away from $\boldsymbol{\gamma}(L)$ in the swarm (the last robot), and let $\mathbf{p}_{\mathrm{s}}$ denote the position of the robot nearest to $\boldsymbol{\gamma}(L)$ in the swarm (the front robot). Let $\mathbf{m}(\mathbf{p}_i)$ denote the projection of the ith robot on the tube generating curve. Then, the position of the front robot in the swarm is

$$\mathbf{p}_{\mathrm{s}} = \arg\min_{\mathbf{p}_i} s(\mathbf{m}(\mathbf{p}_i), \boldsymbol{\gamma}(L)), \tag{9}$$

where $s(\mathbf{m}(\mathbf{p}_i), \boldsymbol{\gamma}(L))$ denotes the arc length of the tube generating curve between $\mathbf{m}(\mathbf{p}_i)$ and $\boldsymbol{\gamma}(L)$. Similarly, the position of the last robot is

$$\mathbf{p}_{\mathrm{e}} = \arg\max_{\mathbf{p}_i} s(\mathbf{m}(\mathbf{p}_i), \boldsymbol{\gamma}(L)). \tag{10}$$

Thus the area occupied by the swarm is

$$S = -\int_{\mathbf{m}(\mathbf{p}_{\mathrm{e}})}^{\mathbf{m}(\mathbf{p}_{\mathrm{s}})} 2\lambda(\mathbf{p})\mathrm{d}s(\mathbf{p}, \boldsymbol{\gamma}(L)). \tag{11}$$

Assumption 1: The swarm is considered as a single point called the swarm center point with average speed v_{a} and swarm density ρ_{a}.

Based on Assumption 1, the swarm center point is depicted as a yellow pentagram in Fig. 4. The forward speed of this

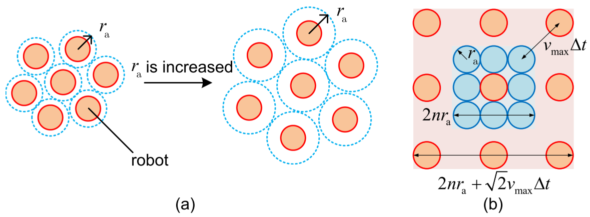


Fig. 5. (a) Swarm density will be decreased if r_{a} is increased. (b) Schematic diagram of calculating the maximum change rate of the swarm density.

point during the passing-through process is planned. The average forward speed of the swarm is $v_{\mathrm{a}}=\sum_{i=1}^{N} v_{\mathrm{f},i}/N$.

E. Problem Formulation

In this letter, a speed-constrained swarm moving in a complex environment is simplified as moving within a virtual tube with varying width $\lambda(l)$. The objective is to enhance the passing-through *safety* and *efficiency*. This virtual tube is pre-designed. Let $v_{\mathrm{a}}^{*}(l)$ and $\rho_{\mathrm{a}}^{*}(l)$ denote the planned average forward speed and the planned density respectively.

Assumption 2: The area occupied by a robot is a circumscribed square of its circular obstacle avoidance area. Moreover, the area occupied by the swarm is minimum when the circumscribed square of robots' circular safety area are closely adjacent.

Assumption 3: Suppose the area occupied by a robot swarm is rectangular. As illustrated in Fig. 5(b), the fastest expansion strategy for the swarm is that the robots at the four corners of the original blue square area occupied by the swarm move away from the center point of the square with the maximum speed $v_{\max}$, and then become the four corners of the new red square area occupied by the swarm.

- *Speed Planning:* Based on Assumption 1, speed planning is employed to determine the average forward speed $v_{\mathrm{a}}^{*}(l)$ of the swarm at each position of the virtual tube for accelerating the passing-through process.
- *Density Planning:* Based on Assumptions 2 and 3, density planning is employed to determine the swarm density $\rho_{\mathrm{a}}^{*}(l)$ at each position of the virtual tube for enhancing the safety of speed-constrained swarms.
- *Tracking Control:* Let the swarm track the planning results of speed and density in the actual passing-through process. The planned average forward speed $v_{\mathrm{a}}^{*}(l)$ is applied directly to $v_{\mathrm{f},i}$ to implement speed tracking. In addition, the avoidance radius $r_{\mathrm{a},i}$ is regulated to modify the velocity components $\mathbf{v}_{\mathrm{m},i}$ and $\mathbf{v}_{\mathrm{xy},i}$. Subsequently, the planned swarm density $\rho_{\mathrm{a}}^{*}(l)$ is tracked.

III. Main Results

A. Speed and Density Planning

In order to enhance the safety of the swarm, the density $\rho_{\mathrm{a}}(l)$ is planned to be as proximate to the desired density ρ_{d} as feasible throughout the whole passing-through process. The constant ρ_{d} is preset from experience as a reasonable value of a collision-free distribution. Additionally, the average forward speed $v_{\mathrm{a}}(l)$ is planned to ensure that the swarm can pass through as fast as possible.

A pre-designed virtual tube is given. According to the analysis above, the following planning is derived.

$$\min_{v_{\mathrm{a}},\rho_{\mathrm{a}}\in C[0,L]} J=\int_0^L \frac{1}{v_{\mathrm{a}}(l)}\mathrm{d}l+\int_0^L\left(\rho_{\mathrm{a}}(l)-\rho_{\mathrm{d}}\right)^2\mathrm{d}l \tag{12}$$

subject to

$$v_{\min} \leq v_{\mathrm{a}}(l)\leq v_{\max}, \tag{13}$$

$$|\dot{v}_{\mathrm{a}}|\leq a_{\mathrm{v}}, \tag{14}$$

$$v_{\mathrm{a}}(l)\leq\sqrt{a_{\mathrm{n}}r_{\mathrm{t}}(l)}, \tag{15}$$

$$0<\rho_{\mathrm{a}}(l)\leq\rho_{\max}, \tag{16}$$

$$|\dot{\rho}_{\mathrm{a}}|\leq a_{\rho}, \tag{17}$$

$$\left(|\rho_{\mathrm{f}}\left(l+\Delta l\right)-\rho_{\mathrm{a}}(l))/\Delta t\right|\leq a_{\rho}, \tag{18}$$

$$a_{\rho}=\sqrt{2}Nv_{\max}\,/\left(4n^3{r_{\mathrm{a}}}^3\right). \tag{19}$$

Here, L, $v_{\min}$, $v_{\max}$, a_{v}, a_{n}, $\rho_{\max}$, N are known constants. The variable $r_{\mathrm{t}}(l)$ denotes the radius of the curvature at the position with the generating curve's arc length l. The constant $\rho_{\max}$ denotes the maximum density permitted for the swarm without collision. The constant a_{ρ} denotes the maximum change rate of the swarm density. The variable $\rho_{\mathrm{f}}(l+\Delta l)$ denotes the predicted density after the swarm moves forward by a distance of Δl along the generating curve without any change in the relative positions of the robots based on $\rho_{\mathrm{a}}(l)$. The constant $n=\lceil\sqrt{N}\rceil$, which denotes rounding up to the closest integer of $\sqrt{N}$.

• Eq. (12) is the objective function. It aims to minimize the passing-through time while ensuring that the density of the swarm approximates the desired density ρ_{d} as closely as possible. Particularly, $\rho_{\mathrm{d}}>0$, an appropriate setting of desired density can trade-off enhance safety and efficiency.

• Constraint (13) restricts the magnitude of the average forward speed $v_{\mathrm{a}}(l)$, which is determined by the physical characteristics of robots according to (2).

• Constraint (14) restricts the change rate of the average forward speed $v_{\mathrm{a}}(l)$, which is determined by the physical characteristics of robots according to (3). Specifically,

$$|\dot{v}_{\mathrm{a}}|=\left|\frac{\mathrm{d}v_{\mathrm{a}}(l)}{\mathrm{d}l}\cdot\frac{\mathrm{d}l}{\mathrm{d}t}\right|=\left|\frac{\mathrm{d}v_{\mathrm{a}}(l)}{\mathrm{d}l}v_{\mathrm{a}}(l)\right|\leq a_{v}. \tag{20}$$

• Constraint (15) restricts the magnitude of $v_{\mathrm{a}}(l)$ at various locations within the virtual tube, which can be derived from (4). Specifically, it is significant to ensure that the swarm does not exceed the boundaries of the virtual tube when it passes through the locations where the tube generating curve is more curved. In other words, if the curvature radius of the tube generating curve is minimal, $v_{\mathrm{a}}(l)$ cannot be excessively high according to the speed constraint of robots in (4).

• Constraint (16) restricts that the swarm density $\rho_{\mathrm{a}}(l)$ may not exceed the maximum density $\rho_{\max}$ for safety. Based on Assumption 2, the minimum area occupied by the swarm is $S_{\min}=4Nr_p^2$, thus the maximum density $\rho_{\max}$ is

$$\rho_{\max}=N/S_{\min}=1/(4r_p^2). \tag{21}$$

• Constraint (17) restricts the change rate of $\rho_{\mathrm{a}}(l)$, thus the area occupied by the swarm in the virtual tube cannot be altered instantaneously. The maximum change rate of the swarm density a_{ρ} is calculated by (19). Specifically,

$$|\dot{\rho}_{\mathrm{a}}|=\left|\frac{\mathrm{d}\rho_{\mathrm{a}}(l)}{\mathrm{d}l}\cdot\frac{\mathrm{d}l}{\mathrm{d}t}\right|=\left|\frac{\mathrm{d}\rho_{\mathrm{a}}(l)}{\mathrm{d}l}v_{\mathrm{a}}(l)\right|\leq a_{\rho}. \tag{22}$$

• Constraint (18) provides predictive density planning according to the pre-designed virtual tube with known parameters. As shown in Fig. 4, $\rho_{\mathrm{f}}(l+\Delta l)$ represents the predicted swarm density after traversing a distance Δl along the tube generating curve without changing the relative positions of any pair of robots from the positions with arc length l of the tube generating curve, where the swarm density is $\rho_{\mathrm{a}}(l)$. The variable $\rho_{\mathrm{f}}(l+\Delta l)$ can be calculated according to (8) and (11), which is related to N and $\lambda(l)$. Specifically,

$$\left|\frac{\rho_{\mathrm{f}}(l+\Delta l)-\rho_{\mathrm{a}}(l)}{\Delta t}\right| = \left|\frac{\rho_{\mathrm{f}}(l+\Delta l)-\rho_{\mathrm{a}}(l)}{\Delta l}v_{\mathrm{a}}(l)\right|$$
$$= f\left(N, \lambda(l), \rho_{\mathrm{a}}(l), v_{\mathrm{a}}(l), \Delta l\right) \le a_{\rho}, \tag{23}$$

where $f(N, \lambda(l), \rho_{\mathrm{a}}(l), v_{\mathrm{a}}(l), \Delta l)$ denotes a function. This formula means that if the swarm moves forward without changing the relative positions of robots, the change rate of the swarm density caused by the variation of the tube width cannot exceed the maximum change rate of the swarm density a_{ρ} calculated by (19). The following planning results demonstrate that this constraint plans a relatively low swarm density before entering the narrowest part of the virtual tube. Consequently, the swarm expands before entering the narrowest part of the virtual tube. In other words, the swarm can compensate for the increase in density caused by the varying tube widths through the active expansion in advance. Therefore, collisions and congestion are avoided before the swarm enters the narrowed part of the virtual tube, thereby enhancing safety and efficiency.

• Constraint (19) indicates that a_{ρ} relies on $N, v_{\max}, r_{\mathrm{a}}$. The value of a_{ρ} is derived as follows. Based on Assumption 3, swarm density changes most rapidly when the swarm expands fastest. As shown in Fig. 5(b), it can be assumed that the robots are located within the blue square with side length of $2nr_{\mathrm{a}}$ initially. After time interval of Δt, the robots expand to the red square with side length of $2nr_{\mathrm{a}}+\sqrt{2}v_{\max}\Delta t$. Therefore, the maximum change rate of the swarm density is

$$a_{\rho} = \left|\lim_{\Delta t\to 0}\frac{\rho_1-\rho_0}{\Delta t}\right| = \left|\lim_{\Delta t\to 0}\frac{\frac{N}{S_1}-\frac{N}{S_0}}{\Delta t}\right| = \frac{\sqrt{2}Nv_{\max}}{4n^3{r_{\mathrm{a}}}^3}, \tag{24}$$

where ρ_0 represents the initial swarm density, and ρ_1 represents the swarm density after time Δt.

B. Tracking Control of Planned Average Forward Speed and Density

The objective of control is to ensure that the swarm adheres to the planned average forward speed $v_{\mathrm{a}}^*(l)$ and swarm density $\rho_{\mathrm{a}}^*(l)$ during the passing-through process.

1) Track Planned Average Forward Speed: In order to make the real-time average forward speed of the swarm track $v_{\mathrm{a}}^*(l)$, the planned average speed $v_{\mathrm{a}}^*(l)$ is directly used as the forward speed component $v_{\mathrm{f},i}$ of the ith robot based on the robot controller (6) as follows:

$$v_{\mathrm{f},i}(l_i) = v_{\mathrm{a}}^*(l_i), l_i \in [0, L]. \tag{25}$$

2) Track Planned Swarm Density: Density tracking is achieved by modifying the avoidance radius r_{a}. In order to ensure that the real-time swarm density $\rho_{\mathrm{r}}(l)$ tracks the planned swarm density $\rho_{\mathrm{a}}^*(l)$ during the passing-through process, the velocity command components $\mathbf{v}_{\mathrm{m},i}$ and $\mathbf{v}_{\mathrm{xy},i}$ of each robot are modified by different setting of $r_{\mathrm{a},i}$, which is a variable. Subsequently, the area occupied by the swarm within the virtual tube is changed. Finally, $\rho_{\mathrm{r}}(l)$ is controlled to be close to $\rho_{\mathrm{a}}^*(l)$ as much as possible. Therefore, the controller for $r_{\mathrm{a},i}$ of the ith robot is designed as follows:

$$r_{\mathrm{ac},i}(l_i) = \begin{cases} r_{\mathrm{a}} & \rho_{\mathrm{r}}(l_i) \le \rho_{\mathrm{a}}^*(l_i) \\ \mathrm{sat2}\left(k_{r_{\mathrm{a}}}\left(\rho_{\mathrm{r}}(l_i)-\rho_{\mathrm{a}}^*(l_i)\right), k_1 r_{\mathrm{a}}\right) & \rho_{\mathrm{r}}(l_i) > \rho_{\mathrm{a}}^*(l_i) \end{cases},$$
$$\mathrm{sat2}(b, a) \triangleq \begin{cases} b & b \le a \\ a & b > a \end{cases}, \tag{26}$$

where $k_{r_{\mathrm{a}}} > 0$, $k_1 > 0$ are coefficients, $a > 0$ is a constant, $l_i \in [0, L]$. According to law (26), when the real-time swarm density $\rho_{\mathrm{r}}(l_i)$ exceeds the planned swarm density $\rho_{\mathrm{a}}^*(l_i)$, the avoidance radius $r_{\mathrm{a},i}$ will be increased. The upper limit of the obstacle avoidance radius is constrained for the robot by applying saturation, thereby mitigating the potential for excessive jitter in the control instructions. Therefore, as illustrated in Fig. 5(a), the real-time swarm density $\rho_{\mathrm{r}}(l_i)$ will be decreased to align with the planned swarm density $\rho_{\mathrm{a}}^*(l_i)$. This expansion decreases the probability of collisions between robots, thereby enhancing the safety of the passing-through process.

Particularly, the density tracking is no longer considered when $\rho_{\mathrm{r}}(l_i) < \rho_{\mathrm{a}}^*(l_i)$ in the tracking controller (26), because there are no safety risks for the swarm according to (16). If higher swarm compactness is required, a *robot cohesion term* [20] can be incorporated into the controller to facilitate density tracking when $\rho_{\mathrm{r}}(l_i) < \rho_{\mathrm{a}}^*(l_i)$, thereby gathering the swarm after traversing a narrow space. Moreover, $\rho_{\mathrm{r}}(l_i)$ can be estimated through LIDAR or visual sensors without communication. Specifically, the ith robot determines the relative positions of nearby robots through perception, thereby ascertaining the number of robots n within its perception range and the area S occupied by them within the virtual tube. Subsequently, $\rho_{\mathrm{r}}(l_i)$ can be calculated by (8). This indicates an implementation of distributed tracking control.

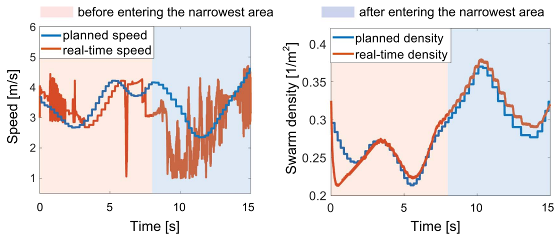
It is important to note that since the design form of the velocity command (6) is consistent with our previous work [7], the tracking controller still maintains stability.

IV. Simulation and Experiment Results

A. Numerical Simulation

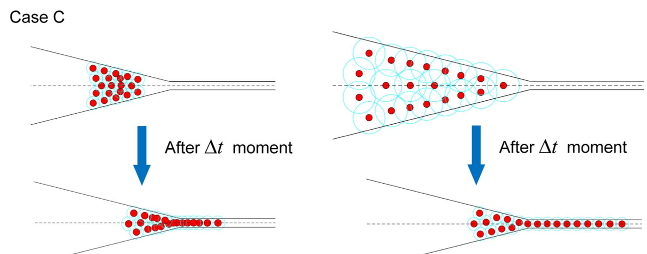
Piecewise polynomials are employed to represent $v_{\mathrm{a}}(l)$ and $\rho_{\mathrm{a}}(l)$ for solving the optimization problem (12). All simulations in MATLAB code are executed on a PC with Intel Core i9-12900KF @ 3.2 GHz CPU and 32G RAM.

1) Simulation With Planning and Without Planning: Simulation comparisons have been conducted between control methods with and without planning across various manually designed virtual tube scenes. Specifically, *control without planning* refers to controlling with default parameters based on our previous control method [7], which refers to (6), while *control with planning* refers to controlling according to the planned average forward speed $v_{\mathrm{a}}^*(l)$ and density $\rho_{\mathrm{a}}^*(l)$ of the swarm. *Virtual tube scenes* include a normally narrowing trapezoidal virtual tube, a normally narrowing curved virtual tube, a rapidly narrowing trapezoidal virtual tube, and a rapidly narrowing curved virtual tube. Parameters are set as follows: $N = 20, v_{\min} = 2$ m/s, $v_{\max} = 5$ m/s, $a_{\mathrm{v}} = 5$ m/s ^{2}, $a_{\mathrm{n}} = 1$ m/s ^{2}, $r_{\mathrm{p}} = 0.3$ m, $r_{\mathrm{s}} = 0.4$ m, $r_{\mathrm{a}} = 0.8$ m, $\rho_{\mathrm{d}} = 0.1989/\mathrm{m}^2$, $\rho_{\max} = 0.9974/\mathrm{m}^2$. In addition, the *passing-through time* is the assessment for *efficiency*, and the *minimum distance* between any pair of robots during the passing-through process is the assessment for *safety*.



(a) Speed tracking (b) Density tracking

Fig. 6. Speed tracking and density tracking in a normally narrowing trapezoidal virtual tube.



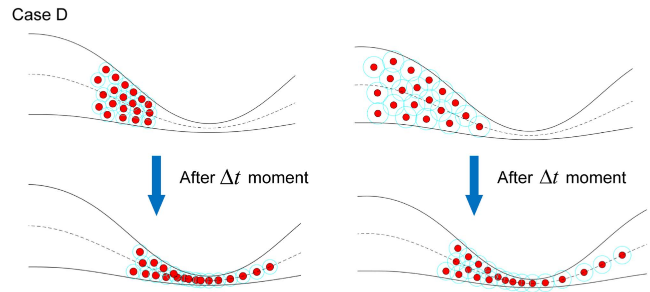
(a) without planning (b) with planning

Fig. 7. Comparison of swarm passing through a rapidly narrowing trapezoidal virtual tube without planning and with planning.

Swarm passing-through time and the minimum distance between robots in four different virtual tube scenes without planning and with planning are recorded in the supplementary materials[1]. The results indicate that the passing-through time with planning is much shorter than without planning. Additionally, the minimum distance between robots with planning is larger than without planning, which demonstrates that speed and density planning enhances the efficiency and safety of the swarm passing-through process effectively.

As depicted in Fig. 6, the swarm can be controlled to track the planning results of speed and density well. In Fig. 6(a), the fluctuations observed in the real-time speed arise due to the employment of the single integrator model according to (1), which permits instantaneous changes in speed. Such fluctuations will not present in practical implementations. In addition, the real-time speed is influenced by both $\mathbf{v}_{\mathrm{m},i}$ and $\mathbf{v}_{\mathrm{xy},i}$. Consequently, the real-time speed generally follows the trend of the planned speed, although not exactly. Particularly, because the real-time swarm density $\rho_{\mathrm{r}}(l)$ is consistently greater than the planned swarm density $\rho_{\mathrm{a}}^{*}(l)$ in Fig. 6, the density tracking is well implemented according to (26). Moreover, it can be found in Fig. 6 that there is a significant decrease in the planned density before entering the narrowest part of the virtual tube. This suggests that the planned density necessitates the swarm to expand before entering the narrowest part, which is consistent with the simulation results. Additionally, it can be observed from Figs. 7 and 8 that there are collisions among robots before entering the narrowest part of the virtual tube without planning. Nevertheless, due to the expansion of the swarm in advance, the collisions are avoided with planning.

[1]https://github.com/RflyBUAA/VirtualTube/blob/main/Speed-and-Density-Planning-for-a-Speed-Constrained-Robot-Swarm-Through-a-Virtual-Tube/SM.pdf.



(a) without planning (b) with planning

Fig. 8. Comparison of swarm passing through a rapidly narrowing curved virtual tube without planning and with planning.

2) Comparative Simulation: In this subsection, the efficacy of the proposed method is evaluated in twenty randomly generated scenes with narrow spaces. Specifically, four methods are compared, including the method proposed in this letter, the basic virtual tube control method [7], the optimized flocking method [21], and the nonlinear model predictive control (NMPC) method [22]. It should be noted that the optimized flocking algorithm is applied to virtual tube control in simulations to illustrate the universality of the virtual tube method simultaneously. Additionally, the virtual tubes in the twenty scenes are automatically generated based on our latest work [17], which demonstrates the possibility of the method proposed being directly applied to swarm navigation in complex environments.

Subsequently, several performance metrics are presented.

- *Arrival time:* The time from when the swarm starts moving until the last robot in the swarm reaches the finish line. If a robot does not reach the finish line within a certain time limit, the arrival time is set to infinity.
- *Arrival rate:* The ratio of the number of robots that safely reach the finish line to the total number of robots. A robot is considered to have safely reached the finish line if it reaches the finish line within a certain time limit without colliding with other robots or with obstacles.
- *Average speed:* The average speed of all robots that reach the finish line within a certain time limit. Robots that do not reach the finish line are not considered.
- *Surpass rate:* The ratio of the number of robots that surpass the virtual tube boundaries to the total number of robots in the swarm. Colliding with the boundary is also considered as surpassing the boundary.

A swarm of twenty robots is controlled to pass through the twenty scenes based on the above four methods. Particularly, considering the applicability of the flocking method and the NMPC method, $v_{\min}$ is set to 0 under these two methods. In other methods, $v_{\min}$ is set to 1, $v_{\max}$ is set to 3. The remaining parameter settings are identical to those in Section IV-A1.

Simulation results are meticulously recorded in the supplementary materials. As a representative example, the simulation results for Scene 2 are presented in Table I and Fig. 9. Taking the *arrival rate* and *arrival time*[2] as indicators to quantify *safety* and *efficiency* respectively, the following conclusions can be drawn: Compared with the basic virtual tube control method, the flocking method and the NMPC method, the proposed method

[2]If the *arrival time* is infinite, it is recorded as 50 seconds when calculating the efficiency improvement ratio.

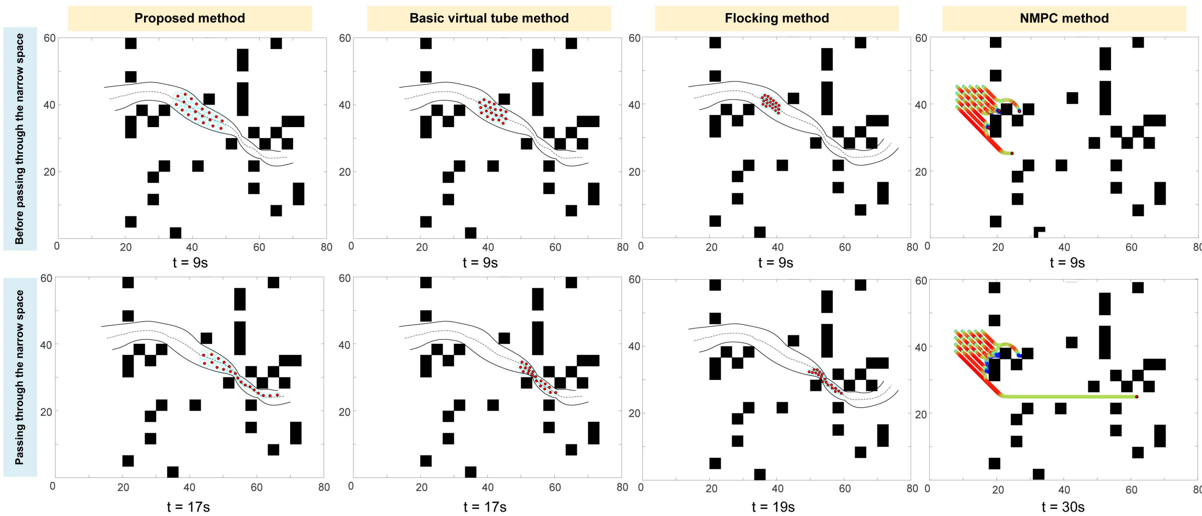


Fig. 9. The passing-through process of the swarm in scene 2 according to the four different methods.

TABLE I
PERFORMANCE METRICS FOR DIFFERENT METHODS IN SCENE 2

Metric	Method			
	PM*	VT†	FM*	NMPC
Arrival time(s)	32.71	**28.13**	46.36	∞
Arrival rate(%)	**100**	30	65	10
Surpass rate(%)	**0**	20	65	\
Average speed(m/s)	2.3269	**2.3734**	2.0769	0.3527

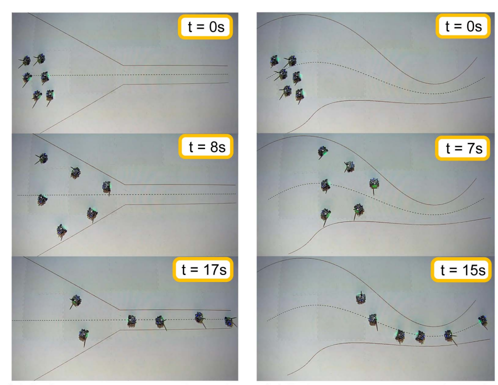
*Proposed method. †Basic virtual tube method. *Flocking method.

is 56.1%, 59.34%, 137.04% safer and 1.71% slower, 12.82%, 19.47% faster respectively. With regard to solving the optimization problem (12), the built-in solver *fmincon* of MATLAB is employed to solve the optimization problems due to the low demand for solving speed in offline planning. Specifically, the conventional Sequential Quadratic Programming (SQP) algorithm for solving nonlinear programming problems is adopted. The solving time is documented in the supplementary materials. Particularly, the average solving time is 257.01 ms.

In conclusion, the method proposed in this letter can enhance the safety and efficiency of the passing-through process in narrow spaces to a large extent. Specially, due to the extreme shape of the virtual tube, the proposed method may sacrifice efficiency to ensure the safety of the swarm.

B. Experiments

1) Experiments on Robotarium: A speed-constrained robot swarm consisting of six robots is employed to conduct experiments on Robotarium[3] [23]. This swarm is required to pass through a narrowing trapezoidal virtual tube and a narrowing curved virtual tube. As shown in Fig. 10, based on the speed and density planning, there are apparent expansions before entering the narrowest part of both virtual tubes around 7 seconds after departure. Consequently, the swarm passes through both virtual tubes without collisions or surpassing the tube boundaries. These



(a) Robot swarm passes through a narrowing trapezoidal virtual tube (b) Robot swarm passes through a narrowing curved virtual tube

Fig. 10. Experiments on Robotarium.

experiments indicate that the proposed method enhances the safety of the passing-through process. It should be noted that a modified controller[4] is designed to better adapt to the time delay of the unicycle model in Robotarium.

2) Experiments on Quadcopters: An experiment based on quadcopters is also conducted in a complex environment with real obstacles. In this experiment, the quadcopters rely on onboard computers to make decisions, achieving a truly distributed control. As shown in Fig. 11, based on an advanced expansion, each quadcopter of the swarm safely passes through the narrowest area where the obstacles are dense. The tracking curves of speed and density during the entire passing-through process are recorded in the supplementary materials, which indicate that the swarm can be controlled to track the planned speed and density

[3]The code is available at https://github.com/RflyBUAA/VirtualTube/tree/main/Speed-and-Density-Planning-for-a-Speed-Constrained-Robot-Swarm-Through-a-Virtual-Tube/Robotarium.

[4]In the modified controller, alterations have been made to the sequence of vector summation and the saturation of control components [24]. The modified controller is formally expressed as: $\mathbf{v}_{\mathrm{c},i} = \mathrm{sat3}(\mathrm{sat3}(\mathbf{v}_{\mathrm{f},i}, v_{\max}) + \mathrm{sat3}(\mathbf{v}_{\mathrm{m},i} + \mathbf{v}_{\mathrm{xy},i}, \|\mathbf{v}_{\mathrm{f},i}\| - v_{\min}), v_{\max})$, where $\mathrm{sat3}(\mathbf{v}, c) \triangleq \begin{cases} \mathbf{v} & \|\mathbf{v}\| \leq c \\ c\mathbf{v}/\|\mathbf{v}\| & \|\mathbf{v}\| > c \end{cases}$, and $c > 0$ is a constant.

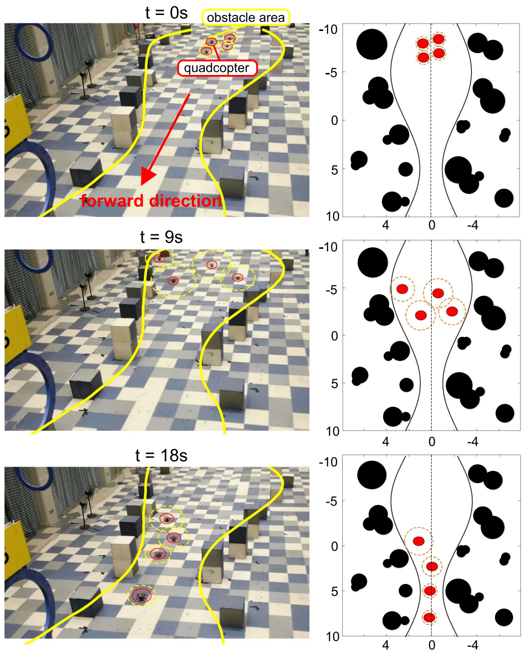


Fig. 11. Flight experiment on quadcopters with real obstacles.

very well. Another experiment of quadcopters is documented in the supplementary materials as well.

V. Conclusion

Speed and density planning with tracking control is proposed in this letter to address the challenge of a speed-constrained robot swarm passing through a known virtual tube with varying widths. The method proposed significantly enhances the safety and efficiency of the passing-through process. It has potential applications in air traffic of drones, a robot swarm traversing a tunnel or searching in a cluttered environment, etc.. Nevertheless, inaccurate robot tracking of the planned speed and density may occasionally result in collisions among robots, which represents a potential area for future study. Influencing factors are as follows. (i) The narrowing degree of the virtual tube. (ii) The limitation of the swarm's ability to expand and track the control command. (iii) The inappropriate settings of the initial states and ideal states. (iv) The length of virtual tubes and the number of robots in the swarm. Furthermore, we intend to integrate the inherent constraints of more accurate robot models directly into speed and density planning. For example, adopting a fixed-wing guidance model will inform the planning of more suitable speed and density profiles for fixed-wing UAVs. Additionally, the implementation of more sophisticated lower-level control strategies is anticipated to achieve enhanced tracking effects, particularly within the constraints imposed by the robot model such as the fixed-wing UAV model.

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